UNIT – I

Productivity and Work Study

Syllabus (as given)

Productivity and Work Study: definition of productivity, task of management, productivity of materials, land, building, machine and power, factors affecting productivity, work content, basic work content, excess work content, how a manufacturing job is made up, work content due to excess product and process, ineffective time due to shortcomings on part of the management.

Definition, objective and scope of Work Study: work study and management, work study and work.

1. Productivity – Definition

- Productivity is the **ratio of output to input.** It tells us how well the resources (men, materials, machines, money, land, energy) are converted into goods / services.
- Productivity = **Output / Input** (output = units produced / sales value ; input = resources consumed)
- European Productivity Agency: "Productivity is an **attitude of mind** – a mentality of progress, of constant improvement of that which exists."
- Productivity does NOT mean working harder. It means working **smarter** – same / more output with same / less input.

Productivity vs Production

Production = total quantity of output (absolute). Productivity = output per unit of input (relative).

Production can rise while productivity falls (if input rises faster than output).

Efficiency = actual output / standard output (how well a target is met); Effectiveness = doing the right thing; Productivity = efficiency + effectiveness together.

Types of productivity

Type	Formula	Remarks
Partial (single factor)	Output / one input (labour, material, capital or energy)	Easy to compute, but may mislead as other inputs are ignored
Multi-factor	Output / (labour + capital + energy ...)	Considers some inputs together

Type	Formula	Remarks
Total factor	Total output (Rs.) / Total input (Rs.)	Best overall indicator; all inputs in money terms

Common partial productivity measures

- Labour productivity = **Output / Man-hours** (units per man-hour)
- Material productivity = Output / Material consumed (kg, Rs.)
- Capital productivity = Output / Capital employed (Rs.)
- Machine productivity = Output / Machine-hours
- Energy productivity = Output / kWh consumed

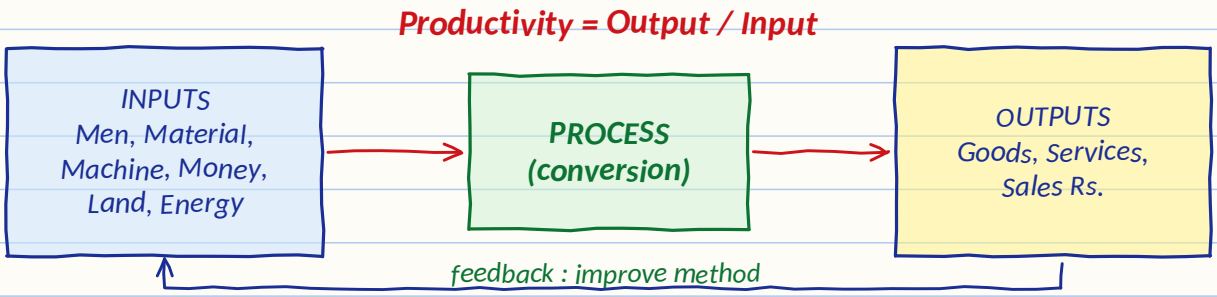


Fig 1.1 Productivity as input–process–output system

Five ways of improving productivity

Case	Output	Input	Result
1	increases ↑	same	Productivity ↑ (most common)
2	same	decreases ↓	Productivity ↑ (cost cutting)
3	increases ↑	decreases ↓	Productivity ↑ ↑ (ideal case, e.g. method study)
4	increases ↑ ↑	increases ↑ (less)	Productivity ↑ (expansion)
5	decreases ↓ (less)	decreases ↓ ↓	Productivity ↑ (retrenchment)

Solved examples

Ex1. A shop makes 4800 units in a month using 1600 man-hours. Labour productivity = 4800 / 1600 = **3 units / man-hour.** After method improvement it makes 5400 units in 1500 man-hours: = 5400 / 1500 = 3.6 units / man-hour. Increase = (3.6 – 3)/3 = **20 %.**

Ex2. Output worth Rs. 6,00,000. Inputs: labour Rs. 1,50,000; materials Rs. 2,50,000; capital Rs. 50,000; energy Rs. 20,000. Total input = Rs. 4,70,000. Total productivity = 6,00,000 / 4,70,000 = **1.277.** Labour productivity = 6,00,000 / 1,50,000 = 4.0.

Ex3. Productivity rises from 2.5 to 2.75 → % increase = (2.75 – 2.5)/2.5 × 100 = **10 %.**

2. Task of Management

- Management's basic task is to **get things done through people** and to make the **best use of the 4 M's** – Men, Materials, Machines, Money (plus Method, Minutes / time).
- Functions: **Planning → Organizing → Staffing → Directing → Controlling** (POSDC).
- To produce goods / services at the **lowest cost,** at the **right quality,** in the **right time** – so that the enterprise survives and grows.
- Management is responsible for:
  - choice of product and its design, specification, standards
  - choice of process, plant, layout, machine and tools
  - purchase and control of materials and stores
  - selection, training and motivation of people, safe conditions
  - planning and controlling production, quality and cost
- Most of the avoidable waste of time and money in a factory (the **ineffective time)** is due to management decisions – hence "the task of management is to reduce the total work content and the ineffective time".

Management cycle (feedback from control to planning)



Fig 1.2 Functions of management

3. Productivity of Materials, Land, Building, Machine & Power

(a) Productivity of materials

- Material productivity = **Output / material input.** Materials are 50–70 % of product cost, so even small saving gives a big gain.
- Ways to improve:
  - proper material selection; cheaper substitute without loss of quality (value analysis / value engineering)
  - reduce scrap, rejects, rework, wastage; increase **yield** (e.g. nesting of blanks, better cutting patterns)
  - standardisation & simplification of materials and parts; bulk purchase, vendor development

- scientific inventory control (EOQ, ABC, JIT), proper storage and handling to avoid damage, obsolescence
- recycling of scrap, use of by-products, proper handling to avoid pilferage
- Yield % = (weight of good product / weight of raw material) × 100

(b) Productivity of land and building

- Land and building are costly and fixed – use them fully. Measure: **Output per sq. m of floor area** (or Rs. per sq. m).
- Improve by: good plant layout; remove unused machines & junk; **vertical storage** / racks; multi-storey buildings; reduce passages and idle space; use of overhead conveyors; proper location of stores.
- Floor-space utilisation = (area occupied by machines, work & aisles in use / total area) × 100.
- Good building: natural light & ventilation, flexible bays, strong floors, easy extension – reduces cost and delays.

(c) Productivity of machines

- Machine productivity = **Output / machine-hours;** Utilisation % = (productive machine time / available time) × 100.
- Improve by: preventive maintenance (reduce breakdown), reduction of set-up time (SMED), correct speeds & feeds, better tooling & fixtures, replacement of obsolete machines, automation, proper machine loading and scheduling, trained operators.
- Overall Equipment Effectiveness: OEE = **Availability × Performance × Quality.**

(d) Productivity of power (energy)

- Energy productivity = **Output / kWh** (or per litre of fuel). Energy cost is a rising share of cost.
- Improve by: power-factor correction (capacitors), energy-efficient motors, switching off idle machines, avoiding peak demand, waste-heat recovery, leak-free compressed air, good insulation and lighting, regular energy audit.

4. Factors Affecting Productivity

- Factors are classified into **internal (controllable)** and **external (uncontrollable)** factors.

Internal factors	External factors
Hard: product, plant & equipment, technology, materials, energy, layout, capital, work methods	Government policies, taxes, tariffs, labour laws

Internal factors	External factors
Soft: people (skill, attitude, motivation), organisation structure, management style, training, wages & incentives, communication, industrial relations	Infrastructure: power, roads, ports, communication
Working conditions: light, noise, temperature, safety, welfare, working hours	Natural resources, climate, geography, social & cultural values, market demand, economy, inflation

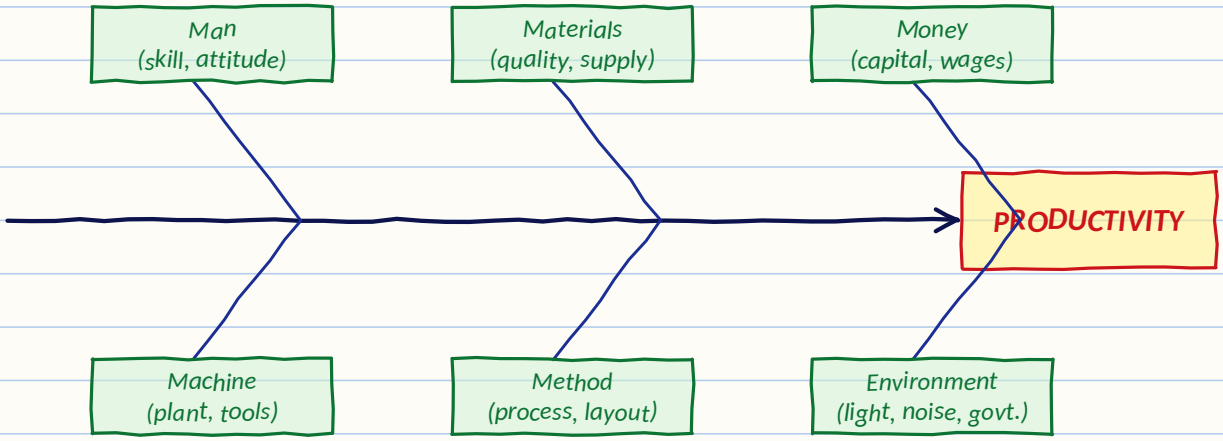


Fig 1.3 Cause-effect (fishbone) diagram of productivity factors

- Other factors: **technology / automation,** quality of work life, **work-study & method improvement,** size of the market, product mix, wage-incentive schemes, TQM, Kaizen, 5S.

5. Work Content

Definition

Work content = the amount of work, measured in man-hours or machine-hours, required to make one unit of a product (to complete the job), when worked at standard performance and with no ineffective time.

Total time to make a product = **Work content + Ineffective time**

Basic work content (BWC)

- The **minimum time theoretically possible** to produce one unit – if the design were perfect, the process perfect and there were no ineffective time of any sort.
- It is the time of the "ideal" product made by the "ideal" process. In reality it is never reached – it is a target to approach (like the speed of light for time!).
- Depends only on the physical / chemical change needed in the material (e.g., the machining time for removing the metal in the most efficient way).

Excess work content (EWC)

- Work content that **adds to the basic work content** because of defects and shortcomings in design, specification or process.
- Two sources:
  - **A.** Work content added by **defects in design / specification of the product**
  - **B.** Work content added by **inefficient methods of manufacture or operation**
- Total work content = Basic work content + A + B

Ineffective time

- **C.** Ineffective time due to **shortcomings of the management**
- **D.** Ineffective time due to **shortcomings of the worker** (within his control)

Equation

Total time of manufacture = Basic work content + A + B + C + D

Total work content = Basic WC + A + B ; Total ineffective time = C + D

6. How a Manufacturing Job is Made Up

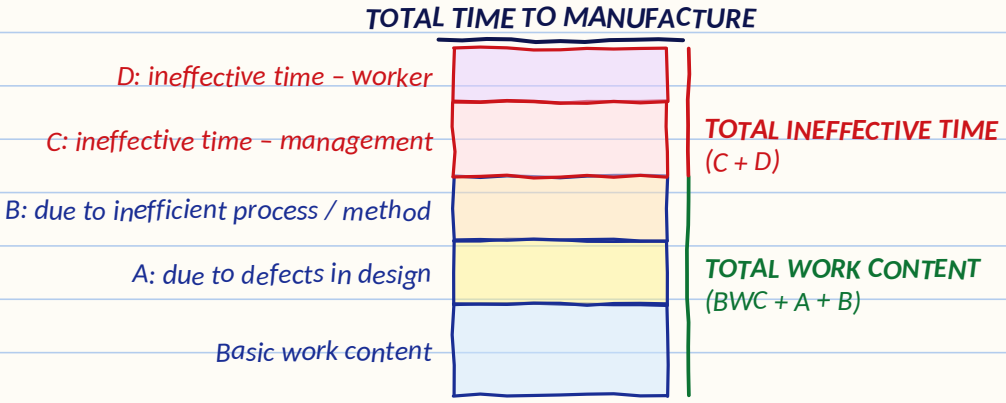


Fig 1.4 Build-up of the total time of a manufacturing job

- The task is to attack the layers from the top: remove D and C first (management action), then B (method study), then A (product redesign) → total time comes down towards the **basic work content.**

Layer	Who can reduce it	Techniques
A (design)	Product designer / R&D / marketing	Value analysis, standardisation, simplification, better specification, design for manufacture
B (process)	Production / process engineer, work-study man	Method study, better tooling, machines, layout, process planning, mechanisation

Layer	Who can reduce it	Techniques
C (management)	Top & middle management	Planning & control, maintenance, stores, training, safe & good conditions, scheduling
D (worker)	Worker with supervisor	Motivation, incentives, discipline, training, joint consultation

7. Work Content due to Excess in Product and Process

(A) Excess work content due to product (design / specification)

- Product design that does not permit the use of the most economical process or method.
- **Lack of standardisation** of parts, materials, sizes – leads to short runs and frequent changeovers.
- Too wide a variety of products / models → small batches, frequent set-ups; products made to individual customer order.
- Unsuitable or too-close tolerances, surface finish, quality standards beyond functional need.
- Too many parts / poor design for manufacture & assembly (DFMA) – more machining, joining, inspection.
- Wrong materials – hard to machine, high waste, heavy; unnecessary stock removal (large machining allowance).
- Frequent design changes; poor liaison between design and production.

(B) Excess work content due to process (method of manufacture)

- **Wrong machines or tools** – e.g. using a lathe where a milling / press tool is faster; worn tools; poor tool design.
- Wrong operating conditions: cutting speed / feed / depth of cut not optimum; wrong coolants.
- Inefficient layout – long transport, back-tracking, waiting between operations; poor material handling.
- Poor sequence and method of work – unnecessary motions, poor workplace layout (no motion economy), poor jigs & fixtures.
- Excess set-up and inspection time, poor process planning, using skilled work for unskilled jobs.
- Poor working conditions – bad light, heat, noise causing fatigue and slow work.

Note: Both A and B can be reduced by method study + value analysis; they are a part of "excess work content".

8. Ineffective Time due to Shortcomings on the Part of Management

- Time during which the **worker is available but nothing useful is produced** because of causes within management control.
- Main causes:
  - **Too wide a variety of products** and frequent changes of design → idle time, rework, set-ups.
  - Failure to standardise materials, parts and methods; bad product design.
  - Poor planning and scheduling → workers wait for work, instructions, drawings, tools, material.
  - Shortage / poor quality of materials (poor purchase & stores control).
  - Machine breakdowns due to poor maintenance; power failure; poor tool supply.
  - Poor working conditions and unsafe practices → accidents, injuries.
  - Lack of training, supervision and motivation; poor industrial relations, strikes.
  - Weak production control, poor layout and material handling → waiting & transport waiting.

Ineffective time due to shortcomings of the worker (D)

- Absenteeism, lateness, idling, slow work (deliberate), careless work causing scrap & rework, ignoring safety rules, taking more rest than permitted.

Note: D is controlled by good supervision, incentive schemes, discipline and training; but the root cause is often a management problem too.

9. Work Study – Definition, Objectives and Scope

Definition (ILO)

Work study is a generic term for those techniques, particularly **method study** and **work measurement,** which are used in the examination of human work in all its contexts, and which lead systematically to the investigation of all the factors which affect the efficiency and economy of the situation being reviewed, in order to effect improvement.

- Work study = **Method study + Work measurement.** Also called "Methods engineering", "Time and Motion study" (older name).
- Aim: to find the **best way of doing a job** and set the **time it should take** – leading to higher productivity with the existing resources (no capital investment).

- Method study → HOW the work is done (reduce work content). Work measurement → HOW LONG it should take (reduce ineffective time).

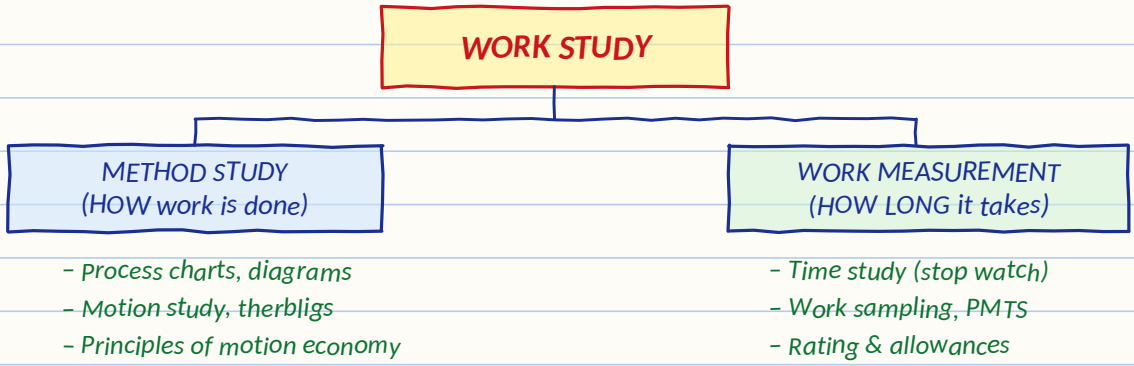


Fig 1.5 Techniques of work study

Objectives of work study

1. To analyse the present method of work and **eliminate unnecessary / wasteful movements** and operations.
2. To develop the **best (standard) method** and standardise it – higher output with less fatigue.
3. To set the **standard time** for the job (for planning, scheduling, costing, wages, incentives).
4. To increase productivity through better use of men, machines, materials and space.
5. To improve plant layout, material handling and working conditions; to reduce fatigue, accidents.
6. To reduce cost, set fair workloads, assess manpower and plan capacity; to give basis for incentive and wage schemes.
7. To train workers in the improved method and to develop co-operation and job satisfaction.

Scope of work study

- Work study is a **universal tool** – applicable wherever human work is done:
  - Manufacturing: machining, assembly, fabrication, inspection, maintenance, stores, packing.
  - Offices & banks (paper work, procedures), hospitals, hotels, retail shops, transport, farms, construction, defence, libraries, software / call centres.
  - Used at all levels: work place, department, whole plant (layout & flow), systems / procedures.
  - Used for new jobs (planning) as well as existing jobs (improvement).

- Why it is attractive: low cost, no or little capital, **systematic,** gives quick results, can be repeated, raises productivity of existing resources.

Basic procedure of work study – "SREDDIM"



- SELECT the job to be studied → RECORD all facts of the present method → EXAMINE critically → DEVELOP the most practical, economic, effective method → DEFINE the new method → INSTALL it as standard practice → MAINTAIN by periodic checks. (Some texts add MEASURE / EVALUATE.)

10. Work Study and Management

- Work study is a **tool of management** – "the servant of management" – used for investigating and improving the use of resources; it is not a replacement for management decisions.
- Management uses work study to:
  - find causes of high work content and ineffective time and remove them;
  - set realistic standards of performance → planning, scheduling, loading, estimating, standard costing, budget;
  - decide manpower, machine requirements and plant layout;
  - control output and cost by comparing actual with standard performance;
  - design wage-incentive plans and fair workloads; support industrial relations.
- Work study gives **facts** to management – it is the "eyes and brain" of the control system. It is a part of industrial engineering and links to quality, maintenance and production control.
- Work study can function effectively only with **top-management support;** the work-study man works as a staff specialist (advises) – the line manager decides and implements.

Work-study man	Line management
Advises, recommends, trains, helps in installation	Takes decisions, implements and maintains the new method
Collects facts, analyses systematically, develops alternatives	Provides resources, authority, follow-up and discipline
Needs tact, patience, integrity, open mind, technical knowledge	Needs to give time, support, consult workers & union

- Qualities of a work-study practitioner: sound technical background, honesty, **tact,** patience, good communication, enthusiasm, ability to inspire confidence, good judgement, physical fitness, sense of humour.

11. Work Study and Work (the Worker)

- Work study studies "human work" – so the worker is the most important partner. Success depends on the worker's **co-operation.**
- Benefits to the worker:
  - easier, less fatiguing, safer way of working; better working conditions;
  - fair standards and wages (incentive earnings), no arbitrary workloads;
  - proper training, job satisfaction, job security through a stronger company;
  - better relations – the standards are based on **measurement and facts,** not on opinion.
- Human problems: fear of job loss / speed-up, fear of change, suspicion of the "stop-watch man", union opposition, loss of skill & status.
- How to win co-operation: explain the aims in advance, consult the workers & union, **involve operators in developing the method,** give credit for ideas, guarantee no retrenchment due to study, behave politely, never criticise the operator, and keep the study open.

Golden rules while studying people

- Inform the supervisor and operator before the study; explain purpose.
- Be honest and impartial in recording; never take "secret" readings.
- Always give credit to the operator for good ideas; never blame him for wastage.
- Share the results with the union & workers; train them in the new method.

Method study vs work measurement – quick comparison

Point	Method study	Work measurement
Aim	Find the best method	Find time for the method
Reduces	Work content (A & B)	Ineffective time (C & D)
Tools	Charts, diagrams, motion study	Stop watch, work sampling, PMTS
Order	Done first	Done after method is standardised

12. Quick Revision & Important Questions

Memory hooks

- Productivity = Output / Input; Types – partial, multi-factor, total.
- Total time = BWC + A (design) + B (process) + C (management) + D (worker).
- Work study = Method study (how) + Work measurement (how long).
- Procedure: SELECT – RECORD – EXAMINE – DEVELOP – DEFINE – INSTALL – MAINTAIN.

Probable exam questions

1. Define productivity. Explain the factors affecting productivity.
2. Distinguish between production and productivity. How can productivity be improved?
3. What is the task of management? Explain how productivity of materials, land and building, machine and power can be raised.
4. Define work content. Explain with a neat sketch how a manufacturing job is made up.
5. Explain basic work content and excess work content; list the causes of excess work content due to product and process.
6. What are the causes of ineffective time due to the shortcomings of management and workers?
7. Define work study. State its objectives and scope. Explain the relation of work study with management and the worker.
8. Explain the basic procedure of work study with a block diagram.
9. Numerical: compute labour productivity, total productivity and % improvement (see Ex1–Ex3).